

The concept of volume, and the volume of a parallelepiped

What a volume is before any formula — three properties, and the cuboid they force.

LESSON 25

1 × 40 MIN

GEOMETRY 11, §12

IGCSE E13.1

LESSON CLOCK

10'

12'

12'

8'

3'

What volume is	10 min
The cuboid	12 min
The parallelepiped	12 min
Units	8 min
Homework	3 min

LEARNING OBJECTIVES

- State the three defining properties of volume.
- Derive $V = abc$ for a cuboid from those properties.
- Compute the volume of a parallelepiped.
- Convert between units of volume.

TEXTBOOK

National	pp. 123–128
Cambridge	Core & Extended, pp. 301–304
Workbook	Exercise 12.1

01

Explanation

The three properties

VOLUME IS DEFINED BY WHAT IT DOES

1 Congruent solids have equal volumes. **2** If a solid is cut into pieces, the volumes add. **3** The unit cube has volume 1.

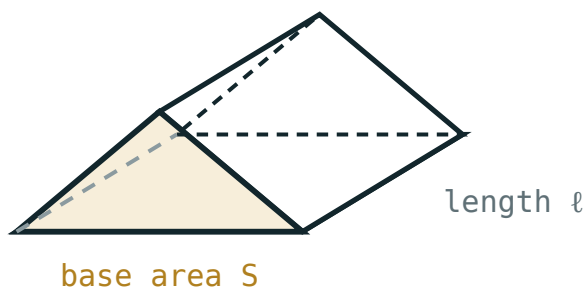
Nothing else is assumed. Every formula in this chapter is a consequence of these three statements, and the first formula follows almost immediately.

The cuboid

A cuboid $a \times b \times c$ with whole-number edges can be filled with exactly abc unit cubes: ab in the bottom layer, and c layers. By properties 2 and 3 its volume is abc .

$$V = abc = B \cdot h$$

For fractional and then irrational edges the same result follows by squeezing between cuboids with rational edges. The formula, once proved for whole numbers, holds for all.



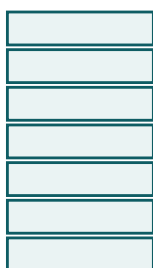
$$V = S \cdot l$$

The bottom layer holds ab cubes; there are c layers.

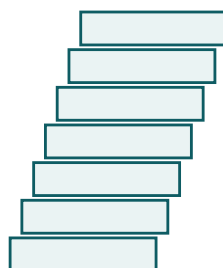
The parallelepiped

An oblique parallelepiped has the same volume as the right one on the same base with the same height. The proof is Cavalieri's principle: every horizontal slice of the two solids has the same area, so the volumes agree.

same base, same height, same volume



right prism



oblique prism

every horizontal slice has the same area

Slide the layers sideways. Nothing is added or removed, so the volume is unchanged.

$V = B \cdot h$ for every parallelepiped, right or oblique

h IS THE PERPENDICULAR HEIGHT

Not the lateral edge. In an oblique solid the lateral edge is longer than the height, and using it inflates the answer.

Units

LENGTH	VOLUME	FACTOR
$1\text{ cm} = 10\text{ mm}$	$1\text{ cm}^3 = 1000\text{ mm}^3$	10^3
$1\text{ m} = 100\text{ cm}$	$1\text{ m}^3 = 1\,000\,000\text{ cm}^3$	100^3
—	$1\text{ litre} = 1000\text{ cm}^3$	—
—	$1\text{ m}^3 = 1000\text{ litres}$	—

CUBE THE LENGTH FACTOR
 1 m^3 is not 100 cm^3 but $1\,000\,000$. This is the k^3 rule of Grade 11 similarity, applied to a change of unit.

02 Worked examples

EXAMPLE 1 A cuboid is $12 \times 8 \times 5\text{ cm}$. Find its volume in cm^3 and in litres.

1 $V = 480\text{ cm}^3$.

2 $1\text{ litre} = 1000\text{ cm}^3$.

3 0.48 litres .

ANSWER
 $480\text{ cm}^3 = 0.48\text{ litres}$

EXAMPLE 2 An oblique parallelepiped has base area 30 cm^2 and lateral edge 10 cm inclined at 60° to the base. Find its volume.

① Height $h = 10 \sin 60^\circ = 5\sqrt{3} \approx 8.66$.

Not 10 .

② $V = 30 \times 8.66$

③ ≈ 259.8

ANSWER

$$150\sqrt{3} \approx 260 \text{ cm}^3$$

EXAMPLE 3 A tank holds 2.4 m^3 . How many litres is that?

① $1 \text{ m}^3 = 1000 \text{ litres}$.

② 2.4×1000

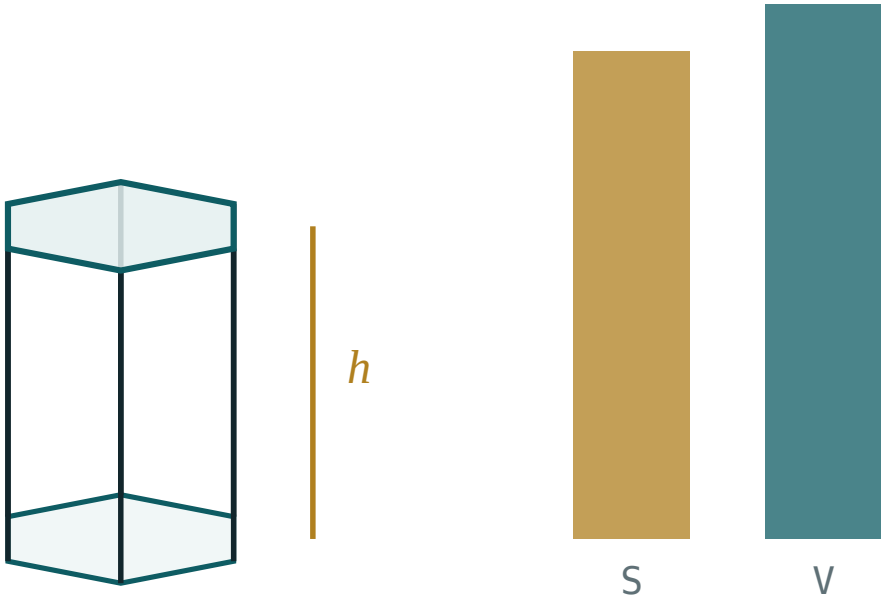
ANSWER

2400 litres

03 Interactive model

Fill a box with unit cubes, one layer at a time, and count aloud.

Base area times height



base area B 41.6

perimeter P 24

total S 227.1

volume V 249.4

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Volume	Hajm	Объём
Unit cube	Birlik kub	Единичный куб
Additivity	Additivlik	Аддитивность
Congruent solids	Teng jismlar	Равные тела
Cubic unit	Kub birlik	Кубическая единица
Parallelepiped	Parallelepiped	Параллелепипед
Base area	Asos yuzasi	Площадь основания
Height of a solid	Jism balandligi	Высота тела
Capacity	Sig'im	Вместимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a unit cube is:

Volume of a cuboid $a \times b \times c$:

1 m³ in cm³:

1 litre in cm^3 :

100

1000

10 000

1

For an oblique solid $V = Bh$ where h is:

the lateral edge

the perpendicular height

the diagonal

the base edge

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a $3 \times 4 \times 5$ cuboid

ANSWER 60

2. Volume of a cube of edge 6

ANSWER 216

3. 1 m^3 in litres

ANSWER 1000

4. 2500 cm^3 in litres

ANSWER 2.5

5. 1 cm^3 in mm^3

ANSWER 1000

6. A cube of volume 125: its edge

ANSWER 5

7. Volume with $B = 20$, $h = 7$

ANSWER 140

Medium · the standard question

7 PROBLEMS

1. Cuboid $12 \times 8 \times 5$ in litres

ANSWER 0.48

2. Oblique parallelepiped, $B = 30$, edge 10 at 60°

ANSWER $150\sqrt{3} \approx 260$

3. A tank of 2.4 m^3 in litres

ANSWER 2400

4. A cuboid of volume 720 with base 12×10 : its height

ANSWER 6

5. A cube of surface area 216: its volume

ANSWER 216

6. Convert 0.75 m^3 to cm^3

ANSWER 750 000

7. A regular hexagonal prism, base edge 4, height 10: volume

ANSWER $240\sqrt{3} \approx 416$

Hard · stretch and reason

7 PROBLEMS

1. A parallelepiped has edges 6, 8, 10 with the 10-edge at 45° to the base. Volume

ANSWER $240\sqrt{2} \approx 339$

2. A cuboid of volume V is scaled by k . New volume

ANSWER k^3V

3. A cube of volume 64 cm^3 : its surface area

ANSWER 96 cm^2

4. A tank $2 \text{ m} \times 1.5 \text{ m} \times 0.8 \text{ m}$ holds how many litres?

ANSWER 2400

5. A brick is $24 \times 12 \times 6 \text{ cm}$. How many in a 1 m^3 wall?

ANSWER 578 (allowing no mortar)

6. Prove $V = abc$ for whole-number edges from the three properties

ANSWER Fill with abc unit cubes and add

7. Why does the same formula hold for irrational edges?

ANSWER Squeeze between rational cuboids

07 Homework

Homework — 4 tasks

Always state the unit, and convert with the cube of the length factor.

- Find the volume of a $15 \times 9 \times 4 \text{ cm}$ cuboid in cm^3 and in litres.
- An oblique parallelepiped has base area 48 cm^2 and lateral edge 12 cm inclined at 30° . Find its volume.
- Convert 3.6 m^3 to litres and to cm^3 .
- Explain in three sentences why 1 m^3 is a million cm^3 and not a hundred.